Ali Assani

Control System Eng.

aliassani19990@gmail.com 05347014548 istanbul Ali Assani

Ali Assani Turkish Male

Profile

A motivated Control and Automation Engineering graduate from Yildiz Technical University with a specialization in control systems and UAV technology. Proficient in MATLAB/Simulink and advanced control algorithms, with hands-on experience in designing and simulating quadcopter systems. Passionate about contributing to cutting-edge aerospace and defense projects, particularly in UAV development, with a strong commitment to innovation and excellence in aircraft control engineering.

Projects

Drone Control Design and Modeling 2024

- Programmed control logic using Arduino IDE (C/C++).
- Designed and fabricated a custom Arduino-compatible PCB shield.
- Integrated gyroscope sensor (MPU6050) for real-time feedback.
- Tuned PID gains manually for optimal flight response.
- Simulated system dynamics and control response in MATLAB.
- Tested and the UAV system in real-time using serial monitor data.

Tools & Technologies: Arduino, PID Control, Custom PCB, MATLAB, Gyroscopes

DJI Mavic Pro Drone simulation 2023

- Modeled linear and angular motion equations of a quadcopter mathematically.
- Implemented PID control systems for altitude and attitude stabilization.
- Simulated full flight dynamics in Simulink using continuous-time blocks.
- Analyzed system behavior under various initial conditions and controller gains.
- Visualized UAV response 3D visualization tools in Simulink.

Quadcopter Simulation 2023

nonlinear

- Developed a nonlinear quadcopter plant model in Simulink based on full motion equations.
- Defined output states as x,y,z (position/altitude), pitch, roll, and Yaw
- Modeled input variables as individual motor thrusts (M1, M2, M3, M4).
- Focused on realistic multi-axis dynamics for future control integration.
- Preparing the system for PID or advanced control implementation (currently under development).

Tools & Technologies: MATLAB, Simulink, Multibody Dynamics, Nonlinear System Modeling

Mini Drone Competition 2025

- Simulated a mini UAV environment with custom control architecture.
- Used Stateflow to model high-level decision-making logic.
- Applied Image Processing Toolbox to detect and respond to visual input data.
- Integrated control and perception layers in a closed-loop simulation.
- Enhanced skills in UAV logic modeling, sensor integration, and mission-level simulation.

Tools & Technologies: MATLAB, Simulink, Stateflow, Image Processing Toolbox

Ball Balancing 2022

- Developed a real-time ball balancing system using Arduino.
- Measured ball position with an HC-SR04 ultrasonic sensor.
- Controlled a servo motor using a PID control algorithm.
- Tuned PID parameters for responsive and stable movement.
- Programmed the system in Arduino IDE (C/C++).
- Physically built and tested the system for performance validation.

Tools & Technologies: Arduino, HC-SR04, Servo Motor, PID Control, Arduino IDE

Education

Yildiz Technical University
2020 – 2025 | istanbul, Türkiye

Skills

- MATLAB/Simulink
- C programming
- Embedded Systems
- problem-solving

Professional Experience

Control and Automation Engineer| Intern 07/2023 – 10/2023 | istanbul, Türkiye

- Gained hands-on experience in PCB design and chip manufacturing processes. Enhancing operational efficiency.
- Collaborated with the engineering team to support the design and optimization of electronic

Courses

Flight Since
MIDDLE EAST TECHNICAL UNIVERSITY

Flight Mechanics
MIDDLE EAST TECHNICAL UNIVERSITY

design of control Systems
YILDIZ TECHNICAL UNIVERSITY

System Dynamics, Modeling and Simulation
YILDIZ TECHNICAL UNIVERSITY

Certificates

- model,simulate and contol a Drone in Matlab and Simulink
- Control Design Onramp With Simulink
- Simulink Onramp
- stateflow onramp
- Model Based Design

Signals and Systems, Applications in Control Engineering
YILDIZ TECHNICAL UNIVERSITY

Real Time Control Systems
YILDIZ TECHNICAL UNIVERSITY

Multivariable Control Theory
Yıldız Tecnichal University

Languages

Turkish

English

Arabic

References

Mücahit Damlahi, *Electronic Engineer,*
Baykar Technology
mojahed.da202@gmail.com, +90 537 741 03 24